National Comprehensive Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Castleman Disease

NCCN Evidence Blocks™

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NCCN recognizes the importance of clinical trials and encourages participation when applicable and available. Trials should be designed to maximize inclusiveness and broad representative enrollment.

NCCN Evidence Blocks™

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NCCN Evidence Blocks™

Find an NCCN Member Institution: https://www.nccn.org/home/member-institutions.

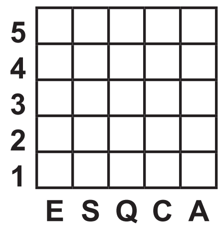
NCCN Categories of Evidence and Consensus: All recommendations are category 2A unless otherwise indicated.

See NCCN Categories of Evidence and Consensus.

NCCN Categories of Preference: All recommendations are considered appropriate.

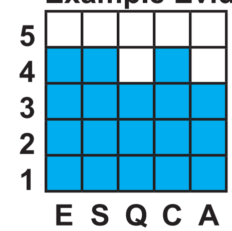
See NCCN Categories of Preference.

NCCN EVIDENCE BLOCKS CATEGORIES AND DEFINITIONS



E = Efficacy of Regimen/Agent
S = Safety of Regimen/Agent
Q = Quality of Evidence
C = Consistency of Evidence
A = Affordability of Regimen/Agent

Example Evidence Block



E = 4
S = 4
Q = 3
C = 4
A = 3

Efficacy of Regimen/Agent

5	**Highly effective:** Cure likely and often provides long-term survival advantage
4	**Very effective:** Cure unlikely but sometimes provides long-term survival advantage
3	**Moderately effective:** Modest impact on survival, but often provides control of disease
2	**Minimally effective:** No, or unknown impact on survival, but sometimes provides control of disease
1	**Palliative:** Provides symptomatic benefit only

Safety of Regimen/Agent

5	**Usually no meaningful toxicity:** Uncommon or minimal toxicities; no interference with activities of daily living (ADLs)
4	**Occasionally toxic:** Rare significant toxicities or low-grade toxicities only; little interference with ADLs
3	**Mildly toxic:** Mild toxicity that interferes with ADLs
2	**Moderately toxic:** Significant toxicities often occur but life threatening/fatal toxicity is uncommon; interference with ADLs is frequent
1	**Highly toxic:** Significant toxicities or life threatening/fatal toxicity occurs often; interference with ADLs is usual and severe

Note: For significant chronic or long-term toxicities, score decreased by 1

Quality of Evidence

5	**High quality:** Multiple well-designed randomized trials and/or meta-analyses
4	**Good quality:** One or more well-designed randomized trials
3	**Average quality:** Low quality randomized trial(s) or well-designed non-randomized trial(s)
2	**Low quality:** Case reports or extensive clinical experience
1	**Poor quality:** Little or no evidence

Consistency of Evidence

5	**Highly consistent:** Multiple trials with similar outcomes
4	**Mainly consistent:** Multiple trials with some variability in outcome
3	**May be consistent:** Few trials or only trials with few patients, whether randomized or not, with some variability in outcome
2	**Inconsistent:** Meaningful differences in direction of outcome between quality trials
1	**Anecdotal evidence only:** Evidence in humans based upon anecdotal experience

Affordability of Regimen/Agent (includes drug cost, supportive care, infusions, toxicity monitoring, management of toxicity)

5	**Very inexpensive**
4	**Inexpensive**
3	**Moderately expensive**
2	**Expensive**
1	**Very expensive**

DIAGNOSIS

- Excisional or incisional biopsy is preferred for the definitive diagnosis.
- A fine-needle aspiration (FNA) biopsy is insufficient for the initial diagnosis or histologic assessment of Castleman disease (CD). A core needle biopsy is not optimal because features of CD are very nonspecific and seeing the whole node with architecture is essential to be confident of the diagnosis. Therefore, core needle biopsy is unlikely to be adequate for definitive pathologic diagnosis of CD.
- Hematopathology review of all slides with at least one paraffin block representative of the lesion. Rebiopsy if consult material is nondiagnostic, preferably the most fluorodeoxyglucose (FDG)-avid, accessible lymph node.

ADDITIONAL DIAGNOSTIC TESTING[a,b,c]

ESSENTIAL:
- Adequate immunophenotyping to establish diagnosis
  - IHC panel: kappa/lambda, CD20, CD3, HHV8 latency-associated nuclear antigen (LANA)

USEFUL UNDER CERTAIN CIRCUMSTANCES:
- Molecular analysis to detect immunoglobulin (Ig) gene rearrangements
- IHC panel: Ki-67 index; Ig heavy chains,[d] CD5, CD10, BCL2, BCL6, cyclin D1, CD21, or CD23, CD38, IRF4/MUM1, PAX5, CD138
- Flow cytometry with peripheral blood and/or biopsy specimen: kappa/lambda, CD19, CD20, CD5, CD23, CD10
- Epstein-Barr virus-encoded RNA in situ hybridization (EBER-ISH)

→ Workup (CD-2)

[a] If a high suspicion of a clonal process remains and other techniques have not resulted in a clear identification of a clonal process, then next-generation sequencing (NGS) can be used.
[b] For HIV-related lymphomas associated with CD, see NCCN Guidelines for B-Cell Lymphomas. For diffuse large B-cell lymphoma (DLBCL) associated with CD in patients without HIV infection, see NCCN Guidelines for B-Cell Lymphomas - DLBCL.
[c] There are three histopathologic subtypes across a spectrum, from hyaline vascular to plasmacytic, with mixed subtypes in between. All histopathologic subtypes can be observed across all subtypes of CD. Fajgenbaum DC, Blood 2017;129:1646-1657.
[d] In HHV8+ plasmacytic variant, plasmablasts are IgM lambda while normal plasma cells are IgG or IgA polytypic.

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WORKUP[e]

ESSENTIAL:
- Physical exam: attention to node-bearing areas, including Waldeyer's ring, and to size of liver and spleen
- Performance status
- Assess for criteria for active disease[f]
- Complete blood count (CBC) with differential
- Comprehensive metabolic panel
- LDH, C-reactive protein (CRP), erythrocyte sedimentation rate (ESR)
- Beta-2-microglobulin
- Serum protein electrophoresis (SPEP) with serum immunofixation electrophoresis (SIFE) and/or quantitative Ig levels, serum free light chain (FLC) assay
- HIV, HHV8, Epstein-Barr virus (EBV) polymerase chain reaction (PCR)
- Hepatitis B testing[g]
- Hepatitis C testing[h]
- Skull-base to mid thigh FDG-PET/CT scan (preferred) or chest/abdomen/pelvis (C/A/P) CT with contrast of diagnostic quality
- Pregnancy testing in patients of childbearing age (if therapy planned)

USEFUL UNDER CERTAIN CIRCUMSTANCES:
- If HHV8 or HIV positive, screening for concurrent Kaposi sarcoma (KS) is strongly recommended
- Bone marrow biopsy + aspirate
- Bone marrow biopsy with trichrome and reticulin stain (particularly in patients with TAFRO syndrome)
- Neck CT with contrast
- Echocardiogram or multigated acquisition (MUGA) scan if anthracycline-based regimen is indicated
- IgG4, sIL6, sIL10, vascular endothelial growth factor receptor (VEGF), uric acid, ferritin[i]
- Discuss fertility preservation[j]

SUBTYPE

Unicentric Castleman disease (UCD) → CD-3

Multicentric Castleman disease (MCD) → CD-4A

[e] See Subtypes of Idiopathic MCD (CD-A). Rule out other diseases that can mimic idiopathic MCD (iMCD) (see Exclusion Criteria on CD-B 1 of 2). If concurrent polyneuropathy and monoclonal plasma cell disorder, a workup for polyneuropathy, organomegaly, endocrinopathy, monoclonal protein, skin changes (POEMS) syndrome is recommended. Patients with POEMS-associated MCD should have treatment directed at POEMS syndrome with regimens recommended for the treatment of multiple myeloma (see NCCN Guidelines for Multiple Myeloma).

[f] Diagnostic Criteria for Idiopathic MCD (CD-B). Active disease is the presence of ≥2 minor criteria (with at least 1 of them being a lab abnormality).

[g] Hepatitis B testing is indicated because of the risk of reactivation with immunotherapy + chemotherapy. See Monoclonal Antibody Therapy and Viral Reactivation (NHODG-B) in the NCCN Guidelines B-Cell Lymphomas. Tests include hepatitis B surface antigen and core antibody for a patient with no risk factors. For patients with risk factors or previous history of hepatitis B, add e-antigen. If positive, check viral load and consider consult with gastroenterologist.

[h] Hepatitis C antibody and if positive, viral load and consult with hepatologist.

[i] Measurement of acute phase reactants may be helpful in monitoring therapy.

[j] Fertility preservation options include: sperm banking, in vitro fertilization (IVF), or ovarian tissue or oocyte cryopreservation.

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FIRST-LINE THERAPY[k] | **RELAPSED DISEASE[k]**

UCD

- **Surgically resectable**
  - **Complete resection** → **Observation** → **Recurrence** → (back to UCD)
  - **Incomplete resection**
    - **Asymptomatic** → **Observation** → **Recurrence** → (back to UCD)
    - **Symptomatic** → **See Surgically unresectable below**
- **Surgically unresectable** → **Involved-site RT (ISRT)[l,m] or Suggested Treatment Regimens: First-line Therapy (CD-C 1 of 3) or Consider embolization or Observation in asymptomatic patients**
  - **Surgically resectable**
    - **Complete resection** → **Observation** → **Relapsed/refractory disease[n]**
    - **Incomplete resection** → **Alternate first-line therapy not previously given (CD-C 1 of 3)** → **Relapsed/refractory disease[n]**
  - **Surgically unresectable** → **Alternate first-line therapy not previously given (CD-C 1 of 3)** → **Relapsed/refractory disease[n]**

Relapsed/refractory disease[n] → **Consider surgery or ISRT[l,m] or embolization if amenable or Suggested Treatment Regimens: Second-line and Subsequent Therapy (CD-C 2 of 3)**

[k] See Supportive Care for B-Cell Lymphomas for the management of complications associated with anti-CD20 monoclonal antibody (mAb) therapy in the NCCN Guidelines for B-Cell Lymphomas.
[l] The optimal dose of ISRT in unresectable UCD is not established. Doses as high as 40 Gy in 1.5–2 Gy fractions have been used (Chronowski GM, et al. Cancer 2001;92:670-676). Lower doses may be considered depending upon clinical circumstances (eg, proximity to critical structures, treatment intent). Advanced radiation techniques (eg, intensity-modulated RT [IMRT], image-guided RT [IGRT], protons) are recommended to limit dose to surrounding normal tissues (Matthiesen C, et al. Radiol Oncol 2012;46:265-270). RT Dose Constraint Guidelines for Lymphoma - Recommendations for normal tissue dose constraints can be found in the Principles of Radiation Therapy section of the NCCN Guidelines for Hodgkin Lymphoma.
[m] Patients with non-bulky disease may be observed after ISRT.
[n] Encourage biopsy to rule out transformation to DLBCL or concomitant development of other malignancies or opportunistic infections.

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FIRST-LINE THERAPY[k]

RELAPSED DISEASE[k]

MCD → **Criteria for active disease[f] present but no organ failure** → **HIV-1(-)/HHV8(-) (Idiopathic MCD [iMCD; Nonsevere][f])** → **Suggested Treatment Regimens: First-line Therapy (CD-C 1 of 3)**

- → **Response[q]** →
  - **If siltuximab, continue until progression**
  - **If rituximab-based time-limited therapy, observe**

  → **Relapsed disease[n]** → **Alternate first-line therapy (CD-C 1 of 3) before moving onto treatment for relapsed/refractory or progressive disease[r]**
- → **No response[n,q]** → **Alternate first-line therapy (CD-C 1 of 3) before moving onto treatment for relapsed/refractory or progressive disease[r]**

→ **Relapsed/Refractory or Progressive Disease (CD-6)**

Criteria for active disease[f] present but no organ failure → **HHV8(+)** → **CD-4B**

MCD → **Fulminant/Severe[o]** → **CD-5**

[f] Diagnostic Criteria for Idiopathic MCD (CD-B). Active disease is the presence of ≥2 minor criteria (with at least 1 of them being a lab abnormality).
[k] See Supportive Care for B-Cell Lymphomas for the management of complications associated with anti-CD20 mAb therapy in the NCCN Guidelines for B-Cell Lymphomas.
[n] Encourage biopsy to rule out transformation to DLBCL or concomitant development of other malignancies or opportunistic infections.
[o] Severe disease defined as: Eastern Cooperative Oncology Group (ECOG) performance status ≥2; stage IV kidney dysfunction (estimated glomerular filtration rate <30 mL/min/1.73 m^2, creatinine >3 mg/dL); anasarca and/or ascites, pleural effusion, or pericardial effusion; hemoglobin ≤8 g/dL; pulmonary involvement/interstitial pneumonitis with dyspnea (van Rhee F, et al. Blood 2018;132:2115-2124).
[p] All patients with HIV infection should be on combination antiretroviral therapy (cART). For principles of concurrent HIV management and supportive care, see the NCCN Guidelines for Cancer in People with HIV.
[q] Response assessment using the imaging modalities performed during workup (C/A/P CT with contrast or PET/CT).
[r] Rituximab ± prednisone may be repeated without limit if progression is ≥6 months after completion of rituximab.

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FIRST-LINE THERAPY[k]

RELAPSED DISEASE[k]

MCD → Criteria for active disease[f] present but no organ failure → HIV-1(+)/HHV8(+)[p] or HIV-1(-)/HHV8(+) → Suggested Treatment Regimens: First-Line Therapy (CD-C 1 of 3)[s]

- Response[q] → Observation → Relapsed disease[n] → Alternate first-line therapy (CD-C 1 of 3) before moving onto treatment for relapsed/refractory or progressive disease[r] → Relapsed/Refractory or Progressive Disease (CD-6)
- No response[n,q] → Alternate first-line therapy (CD-C 1 of 3) before moving onto treatment for relapsed/refractory or progressive disease[r] → Relapsed/Refractory or Progressive Disease (CD-6)

[f] Diagnostic Criteria for Idiopathic MCD (CD-B). Active disease is the presence of ≥2 minor criteria (with at least 1 of them being a lab abnormality).
[k] See Supportive Care for B-Cell Lymphomas for the management of complications associated with anti-CD20 mAb therapy in the NCCN Guidelines for B-Cell Lymphomas.
[n] Encourage biopsy to rule out transformation to DLBCL or concomitant development of other malignancies or opportunistic infections.
[p] All patients with HIV infection should be on cART. For principles of concurrent HIV management and supportive care, see the NCCN Guidelines for Cancer in People with HIV.
[q] Response assessment using the imaging modalities performed during workup (C/A/P CT with contrast or PET/CT).
[r] Rituximab ± prednisone may be repeated without limit if progression is ≥6 months after completion of rituximab.
[s] Occult KS is prevalent in HIV/HHV8+ MCD and is likely to flare after rituximab or prednisone. Consider baseline imaging and direct visualization to screen for pulmonary ± gastrointestinal (GI) involvement. Monitor closely for KS progression and continue KS-directed therapy for ~3 months post last rituximab to minimize risk of KS flare (given long action of rituximab). See NCCN Guidelines for Kaposi Sarcoma.

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FIRST-LINE THERAPY[k]

RELAPSED DISEASE[k]

Fulminant/Severe[o] HHV8(+) MCD ± organ failure or Fulminant/ Severe[o] HHV8(-) iMCD[t] → **Suggested Treatment Regimens: First-Line Therapy (CD-C 1 of 3)[s]**

- **Response[q]** → **Observation** → **Relapsed disease[n]** → **Relapsed/ Refractory or Progressive Disease (CD-6)**
- **No response[n,q]** → **Relapsed/ Refractory or Progressive Disease (CD-6)**

[k] See Supportive Care for B-Cell Lymphomas for the management of complications associated with anti-CD20 mAb therapy in the NCCN Guidelines for B-Cell Lymphomas.
[n] Encourage biopsy to rule out transformation to DLBCL or concomitant development of other malignancies or opportunistic infections.
[o] Severe disease defined as: ECOG performance status ≥2; stage IV kidney dysfunction (estimated glomerular filtration rate <30 mL/min/1.73 m^2, creatinine >3 mg/dL); anasarca and/or ascites, pleural effusion, or pericardial effusion; hemoglobin ≤8 g/dL; pulmonary involvement/interstitial pneumonitis with dyspnea (van Rhee F, et al. Blood 2018;132:2115-2124).
[q] Response assessment using the imaging modalities performed during workup (C/A/P CT with contrast or PET/CT).
[s] Occult KS is prevalent in HIV/HHV8+ MCD and is likely to flare after rituximab or prednisone. Consider baseline imaging and direct visualization to screen for pulmonary ± GI involvement. Monitor closely for KS progression and continue KS-directed therapy for ~3 months post last rituximab to minimize risk of KS flare (given long action of rituximab). See NCCN Guidelines for Kaposi Sarcoma.
[t] Siltuximab is the preferred treatment option. Monitor organ function. Early intervention with chemoimmunotherapy is recommended for patients with progressive organ dysfunction.

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RELAPSED/REFRACTORY OR PROGRESSIVE DISEASE (MCD)[k]

Relapsed/Refractory or progressive disease (MCD)[n,s]

- **HHV8(+) - Criteria for active disease present with no organ failure or Fulminant/severe[o] [HHV8(+) or HHV8(-)] disease with no organ failure** → **Suggested Treatment Regimens: Second-Line Therapy (CD-C 2 of 3)[u]**
- **Fulminant/severe[o] [HHV8(+) or HHV8(-)] disease with organ failure** → **Suggested Treatment Regimens: Second-Line Therapy (CD-C 2 of 3)[u]**

→ **Response[q]** → **Observation or Maintenance valganciclovir if HHV8(+)** → **Relapsed/refractory disease[n]**

→ **No response[n,q]** → **Alternate second-line combination therapy ± rituximab (CD-C 2 of 3) not previously given before moving onto subsequent therapy** → **Relapsed/refractory disease[n]**

Relapsed/refractory disease[n] → **Consider alternative single-agent or combination therapy - Suggested Treatment Regimens: Alternative Regimens for Subsequent Therapy (CD-C 2 of 3)**

[k] See Supportive Care for B-Cell Lymphomas for the management of complications associated with anti-CD20 mAb therapy in the NCCN Guidelines for B-Cell Lymphomas.
[n] Encourage biopsy to rule out transformation to DLBCL or concomitant development of other malignancies or opportunistic infections.
[o] Severe disease defined as: ECOG performance status ≥2; stage IV kidney dysfunction (estimated glomerular filtration rate <30 mL/min/1.73 m^2, creatinine >3 mg/dL); anasarca and/or ascites, pleural effusion, or pericardial effusion; hemoglobin ≤8 g/dL; pulmonary involvement/interstitial pneumonitis with dyspnea (van Rhee F, et al. Blood 2018;132:2115-2124).
[q] Response assessment using the imaging modalities performed during workup (C/A/P CT with contrast or PET/CT).
[s] Occult KS is prevalent in HIV/HHV8+ MCD and is likely to flare after rituximab or prednisone. Consider baseline imaging and direct visualization to screen for pulmonary ± GI involvement. Monitor closely for KS progression and continue KS-directed therapy for ~3 months post last rituximab to minimize risk of KS flare (given long action of rituximab). See NCCN Guidelines for Kaposi Sarcoma.
[u] Single-agent therapy is preferred for asymptomatic patients with no organ failure; combination therapy is preferred for patients with fulminant disease and organ failure.

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SUBTYPES OF IDIOPATHIC MCD
• Idiopathic MCD (iMCD-TAFRO)[a] ▸ Marked inflammatory syndrome ▸ Thrombocytopenia, anasarca, fever/elevated C-reactive protein (CRP), renal dysfunction/reticulin myelofibrosis, organomegaly ▸ Megakaryocytic hyperplasia, hypervascular or mixed histopathology ▸ Normal immunoglobulin levels • Idiopathic MCD - not otherwise specified (iMCD-NOS)[a] ▸ Less intense inflammatory syndrome ▸ Normal/elevated platelet counts ▸ Plasmacytic or mixed histopathology ▸ Polyclonal hypergammaglobulinemia • Idiopathic plasmacytic lymphadenopathy (IPL)[b] ▸ Thrombocytosis ▸ Polyclonal hyperglobulinemia ▸ Less severe fluid accumulation

[a] Iwaki N, Fajgenbaum DC, Nabel CS, et al. Clinicopathologic analysis of TAFRO syndrome demonstrates a distinct subtype of HHV-8-negative multicentric Castleman disease. Am J Hematol 2016;91:220-226.
[b] Fajgenbaum DC, Uldrick TS, Bagg A, et al. International, evidence-based consensus diagnostic criteria for HHV-8-negative/idiopathic multicentric Castleman disease. Blood. 2017;129:1646-1657.

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Consensus Diagnostic Criteria for Idiopathic MCD[a]		
I. Major Criteria (need both):	1. Histopathologic lymph node features consistent with the iMCD spectrum. Features along the iMCD spectrum include (need grade 2–3 for either regressive GCs or plasmacytosis at minimum): - Regressed/atrophic/atretic germinal centers, often with expanded mantle zones composed of concentric rings of lymphocytes in an “onion skinning” appearance - FDC prominence - Vascularity, often with prominent endothelium in the interfollicular space and vessels penetrating into the GCs with a “lollipop” appearance - Sheetlike, polytypic plasmacytosis in the interfollicular space - Hyperplastic GCs 2. Enlarged lymph nodes (≥1 cm in short-axis diameter) in ≥2 lymph node stations	
II. Minor Criteria (need at least 2 of 11 criteria with at least 1 laboratory criterion)	Laboratory* 1. Elevated CRP (>10 mg/L) or ESR (>15 mm/h)† 2. Anemia (hemoglobin <12.5 g/dL for males, hemoglobin <11.5 g/dL for females) 3. Thrombocytopenia (platelet count <150 μ/mL) or thrombocytosis (platelet count >400 μ/mL) 4. Hypoalbuminemia (albumin <3.5 g/dL) 5. Renal dysfunction (eGFR <60 mL/min/1.73m^2) or proteinuria (total protein 150 mg/24 h or 10 mg/100 mL) 6. Polyclonal hypergammaglobulinemia (total γ globulin or immunoglobulin G >1700 mg/dL)	Clinical 1. Constitutional symptoms: night sweats, fever (>38°C), weight loss, or fatigue (≥2 CTCAE lymphoma score for B-symptoms) 2. Large spleen and/or liver 3. Fluid accumulation: edema, anasarca, ascites, or pleural effusion 4. Eruptive cherry hemangiomatosis or violaceous papules 5. Lymphocytic interstitial pneumonitis
III. Exclusion Criteria (must rule out each of these diseases that can mimic iMCD)	Infection-related disorders 1. HHV-8 (infection can be documented by blood PCR; diagnosis of HHV-8–associated MCD requires positive LANA-1 staining by IHC, which excludes iMCD) 2. Clinical EBV-lymphoproliferative disorders such as infectious mononucleosis or chronic active EBV (detectable EBV viral load not necessarily exclusionary) 3. Inflammation and adenopathy caused by other uncontrolled infections (eg, acute or uncontrolled CMV, toxoplasmosis, HIV, active tuberculosis)	Autoimmune/autoinflammatory diseases (requires full clinical criteria; detection of autoimmune antibodies alone is not exclusionary) 1. Systemic lupus erythematosus 2. Rheumatoid arthritis 3. Adult-onset Still disease 4. Juvenile idiopathic arthritis 5. Autoimmune lymphoproliferative syndrome Malignant/lymphoproliferative disorders (these disorders must be diagnosed before or at the same time as iMCD to be exclusionary): 1. Lymphoma (Hodgkin and non-Hodgkin) 2. Multiple myeloma 3. Primary lymph node plasmacytoma 4. FDC sarcoma 5. POEMS syndrome‡

[a] Fajgenbaum DC, Uldrick TS, Bagg A, et al. International, evidence-based consensus diagnostic criteria for HHV-8-negative/idiopathic multicentric Castleman disease. Blood 2017;129:1646-1657.

Continued

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Consensus Diagnostic Criteria for Idiopathic MCD (continued)[a]

- Select additional features supportive of, but not required for diagnosis
  - Elevated IL-6, sIL-2R, VEGF, IgA, IgE, LDH, and/or B2M
  - Reticulin fibrosis of bone marrow (particularly in patients with TAFRO syndrome)
  - Diagnosis of disorders that have been associated with iMCD: paraneoplastic pemphigus, bronchiolitis obliterans organizing pneumonia, autoimmune cytopenias, polyneuropathy (without diagnosing POEMS‡), glomerular nephropathy, inflammatory myofibroblastic tumor

[a] Fajgenbaum DC, Uldrick TS, Bagg A, et al. International, evidence-based consensus diagnostic criteria for HHV-8-negative/idiopathic multicentric Castleman disease. Blood 2017;129:1646-1657.

Footnotes for CD-B 1 of 2

* Laboratory cutoff thresholds are provided as guidance. Since some laboratories have slightly different ranges, the upper and lower ranges from a particular laboratory should be used to determine if a patient meets a particular laboratory Minor Criterion.

† Evaluation of CRP is mandatory and tracking CRP levels is highly recommended, but ESR will be accepted if CRP is not available.

‡ POEMS is considered to be a disease "associated" with CD. Because the monoclonal plasma cells are believed to drive the cytokine storm, we do not consider it iMCD, but rather "POEMS-associated MCD."

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SUGGESTED TREATMENT REGIMENS[a]

FIRST-LINE THERAPY		
UCD (surgically unresectable/incomplete resection)	**MCD - Criteria for active disease present with no organ failure**	**MCD - Fulminant/severe ± organ failure**
• Rituximab ± Prednisone (for compressive symptoms) • Siltuximab (for primarily inflammatory symptoms) (if HIV-1-negative/HHV8-negative) • Tocilizumab (for primarily inflammatory symptoms) (if HIV-1-negative/HHV8-negative)	***HHV8-negative/HIV-1-negative (iMCD-Nonsevere)*** • Siltuximab • Tocilizumab • Rituximab ± Prednisone • CVP (Cyclophosphamide/Vincristine/Prednisone) + Rituximab	***Fulminant/severe HHV8-negative (iMCD)*** **Preferred** • Siltuximab[b] **Other Recommended** • Tocilizumab • CHOP (Cyclophosphamide/Doxorubicin/Vincristine/Prednisone) ± Rituximab • CVAD (Cyclophosphamide/Vincristine/ Doxorubicin/Dexamethasone) ± Rituximab • CVP ± Rituximab • If not a candidate for combination therapy: ▸ Rituximab
	HHV8-positive (HIV-1-positive or HIV-1-negative)[c] • Rituximab • Rituximab + Liposomal Doxorubicin • Rituximab + Prednisone • Rituximab + Prednisone + Liposomal Doxorubicin • High-dose Zidovudine/Ganciclovir or Valganciclovir	***Fulminant/severe HHV8-positive MCD[c]*** • Rituximab + Liposomal Doxorubicin • Rituximab + Prednisone + Liposomal Doxorubicin • CHOP + Rituximab • CVAD + Rituximab • CVP + Rituximab • If not a candidate for combination therapy: ▸ Liposomal Doxorubicin ▸ Rituximab

See Evidence Blocks on EB-1

[a] An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines. Rituximab and hyaluronidase human injection for subcutaneous use may be substituted for rituximab after patients have received the first full dose of rituximab by intravenous infusion.
[b] Monitor organ function. If stable, continue siltuximab. Early intervention with chemoimmunotherapy is recommended for patients with progressive organ dysfunction.
[c] Occult KS is prevalent in HIV/HHV8+ MCD and is likely to flare after rituximab or prednisone. Consider baseline imaging and direct visualization to screen for pulmonary ± gastrointestinal (GI) involvement. Monitor closely for KS progression and continue KS-directed therapy for ~3 months post last rituximab to minimize risk of KS flare (given long action of rituximab). See NCCN Guidelines for Kaposi Sarcoma.

References on CD-C 3 of 3

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF
All recommendations are category 2A unless otherwise indicated.

SUGGESTED TREATMENT REGIMENS[a]

SECOND-LINE AND SUBSEQUENT THERAPY FOR RELAPSED/REFRACTORY OR PROGRESSIVE DISEASE (in alphabetical order)		
UCD (surgically unresectable/ incomplete resection)	**MCD[d]**	
	HHV8(+) - Criteria for active disease present with no organ failure or Fulminant/Severe disease [HHV8(+) or HHV8(-)] with no organ failure	**Fulminant/Severe disease [HHV8(+) or HHV8(-)] with organ failure**
Treat with alternative first-line therapy regimens (CD-C 1 of 3) before moving onto second-line and subsequent therapy for relapsed/refractory or progressive disease • Rituximab ± Prednisone • Siltuximab[e] (if HIV-1-negative/ HHV8-negative)	**For active disease present with no organ failure, treat with alternative first-line therapy regimens (CD-C 1 of 3) before moving onto second-line and subsequent therapy for relapsed/refractory or progressive disease** **Preferred** (Add ganciclovir or valganciclovir, if HHV8-positive) • Etoposide (oral or IV) • Liposomal Doxorubicin (if HHV8-positive) • Siltuximab[e] (if HHV8-negative) • Vinblastine **Other Recommended** • Sirolimus (if HHV8-negative) • If not previously given ‣ CHOP ± Rituximab ‣ CVAD ± Rituximab ‣ CVP ± Rituximab	**Treat with alternate second-line combination therapy regimens ± rituximab (not previously given) before moving onto subsequent therapy with alternative regimens for relapsed/refractory or progressive disease** **Preferred** • If not previously given ‣ CHOP ± Rituximab ‣ CVAD ± Rituximab ‣ CVP ± Rituximab ‣ Liposomal Doxorubicin ± Rituximab (if HHV8-positive) ‣ Siltuximab[e] (if HHV8-negative)
	Alternative regimens for subsequent therapy: • Anakinra • Bortezomib ± Rituximab • Lenalidomide ± Rituximab • Thalidomide ± Rituximab • Tocilizumab • High-Dose Zidovudine/Ganciclovir or Valganciclovir (if HHV8-positive)	See Evidence Blocks on EB-2 and EB-3

[a] An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines. Rituximab and hyaluronidase human injection for subcutaneous use may be substituted for rituximab after patients have received the first full dose of rituximab by intravenous infusion.
[d] Maintenance therapy (optional) for patients with HHV8-positive disease responding to second-line and subsequent therapy.
[e] Tocilizumab is an appropriate option, if there is a shortage of siltuximab or if it is not available.

References on CD-C 3 of 3

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF
All recommendations are category 2A unless otherwise indicated.

SUGGESTED TREATMENT REGIMENS
REFERENCES

Unicentric Castleman Disease

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van Rhee F, Oksenhendler E, Srkalovic G, et al. International evidence-based consensus diagnostic and treatment guidelines for unicentric Castleman disease. Blood Adv 2020;4:6039-6050.

Multicentric Castleman Disease

Gerard L, Berezne A, Galicier L, et al. Prospective study of rituximab in chemotherapy-dependent human immunodeficiency virus associated multicentric Castleman's disease: ANRS 117 Castleman B Trial. J Clin Oncol 2007;25:3350-3356.

Uldrick TS, Polizzotto MN, Aleman K, et al. High-dose zidovudine plus valganciclovir for Kaposi sarcoma herpesvirus-associated multicentric Castleman disease: a pilot study of virus-activated cytotoxic therapy. Blood 2011;117:6977-6986.

Ramasamy K, Gandhi S, Tenant-Flowers M, et al. Rituximab and thalidomide combination therapy for Castleman disease. Br J Haematol 2012;158:421-423.

Uldrick TS, Polizzotto MN, Aleman K, et al. Rituximab plus liposomal doxorubicin in HIV-infected patients with KSHV-associated multicentric Castleman disease. Blood 2014;124:3544-3552.

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van Rhee F, Rosenthal A, Kanhai K, et al. Siltuximab is associated with improved progression-free survival in idiopathic multicentric Castleman disease. Blood Adv 2022;6:4773-4781.

Pierson SK, Lim MS, Srkalovic G, et al. Treatment consistent with idiopathic multicentric Castleman disease guidelines is associated with improved outcomes. Blood Adv 2023;7:6652-6664.

Liu YT, Gao YH, Zhao H, et al. Sirolimus is effective for refractory/relapsed idiopathic multicentric Castleman disease: A single-center, retrospective study. Ann Hematol 2024;103:4223-4230.

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF
All recommendations are category 2A unless otherwise indicated.

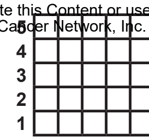
EVIDENCE BLOCKS FOR UCD and MCD

UCD (SURGICALLY UNRESECTABLE/ INCOMPLETE RESECTION)	First-line therapy	Second-line and subsequent therapy for R/R disease
Rituximab	E3 S4 Q3 C3 A3	E3 S4 Q3 C3 A3
Rituximab + Prednisone	E3 S4 Q3 C3 A3	E3 S4 Q3 C3 A3
Siltuximab, HIV-1-negative/HHV8-negative	E4 S4 Q4 C4 A2	E3 S4 Q3 C3 A2
Tocilizumab, HIV-1-negative/ HHV8-negative	E3 S4 Q3 C3 A2	—

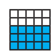
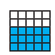
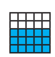
MCD, HHV8-negative, idiopathic MCD: First-line therapy		
	Non-severe criteria for active disease present/ no organ failure	Fulminant/ severe +/- organ failure
CHOP	—	E3 S2 Q3 C3 A3
CVAD	—	E3 S2 Q3 C3 A3
CVP	—	E3 S2 Q3 C3 A3
Rituximab	E3 S4 Q3 C3 A3	E3 S4 Q3 C3 A3
Rituximab + Prednisone	E3 S4 Q3 C3 A3	—
CHOP + Rituximab	—	E3 S2 Q3 C3 A3
CVAD + Rituximab	—	E3 S2 Q3 C3 A3
CVP + Rituximab	E3 S3 Q3 C3 A3	E3 S3 Q3 C3 A3
Siltuximab	E3 S4 Q3 C3 A2	E3 S4 Q3 C3 A2
Tocilizumab	E3 S4 Q3 C3 A2	E3 S4 Q3 C3 A2

MCD, HHV8-positive: First-line therapy		
	Criteria for active disease present/ no organ failure	Fulminant/severe +/- organ failure
Rituximab	E3 S4 Q3 C3 A3	E3 S3 Q3 C3 A3
Rituximab + Liposomal Doxorubicin	E3 S3 Q3 C3 A3	E3 S3 Q3 C3 A3
Rituximab + Prednisone	E3 S4 Q3 C3 A3	—
Rituximab + Prednisone + Liposomal Doxorubicin	E3 S3 Q3 C3 A3	E3 S2 Q3 C3 A3
Liposomal Doxorubicin	—	E3 S3 Q3 C3 A3
CHOP + Rituximab	—	E3 S2 Q3 C3 A3
CVAD + Rituximab	—	E3 S2 Q3 C3 A3
CVP + Rituximab	—	E3 S3 Q3 C3 A3
High-Dose Zidovudine/ Ganciclovir	E3 S3 Q3 C3 A3	—
High-Dose Zidovudine/ Valganciclovir	E3 S3 Q3 C3 A3	—

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF

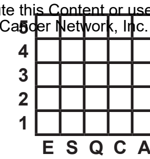


EVIDENCE BLOCKS FOR MCD

MCD: Second-line and subsequent therapy for R/R or progressive disease		
	HHV8(+) criteria for active disease present/no organ failure or Fulm/Severe disease (HHV8 +/-) with no organ failure	HHV8 +/- Fulminant/severe with organ failure
Anakinra		
Bortezomib		
Bortezomib + Rituximab		
CHOP		
CVAD		
CVP		
Etoposide (HHV8-negative)		—
Etoposide/Ganciclovir (HHV8-positive)		—
Etoposide/Valganciclovir (HHV8-positive)		—
Lenalidomide		
Lenalidomide + Rituximab		
Liposomal Doxorubicin (HHV8-positive)	—	
Liposomal Doxorubicin/Ganciclovir (HHV8-positive)		—
Liposomal Doxorubicin + Rituximab (HHV8-positive)	—	
Liposomal Doxorubicin/Valganciclovir (HHV8-positive)		—

[(continued)](#)

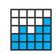
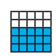
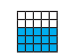
Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#)



E = Efficacy of Regimen/Agent
S = Safety of Regimen/Agent
Q = Quality of Evidence
C = Consistency of Evidence
A = Affordability of Regimen/Agent

NCCN Guidelines Index
Table of Contents
Discussion

EVIDENCE BLOCKS FOR MCD

MCD: Second-line and subsequent therapy for R/R or progressive disease		
	HHV8(+) criteria for active disease present/no organ failure or Fulm/Severe disease (HHV8 +/-) with no organ failure	HHV8 +/- Fulminant/severe with organ failure
CHOP + Rituximab		
CVAD + Rituximab		
CVP + Rituximab		
Siltuximab (HHV8-negative)		
Sirolimus (HHV8-negative)		—
Thalidomide		
Thalidomide + Rituximab		
Tocilizumab		
Vinblastine (HHV8-negative)		—
Vinblastine/Ganciclovir (HHV8-positive)		—
Vinblastine/Valganciclovir (HHV8-positive)		—
High-Dose Zidovudine/Valganciclovir (HHV8-positive)		
High-Dose Zidovudine/Ganciclovir (HHV8-positive)		

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF

Classification

Table 1

The International Consensus Classification (ICC) of Mature Lymphoid Neoplasms (2022)
Mature B-cell lymphomas
HHV-8–associated lymphoproliferative disorders
Multicentric Castleman disease
Primary effusion lymphoma
HHV8-positive diffuse large B-cell lymphoma, NOS
HHV8-positive germinotropic lymphoproliferative disorder

Table 2

WHO Classification of Hematolymphoid Tumors: Lymphoid Neoplasms (2022; 5th edition)
Tumour-like lesions with B-cell predominance
Reactive B-cell-rich lymphoid proliferations that can mimic lymphoma
IgG4-related disease
Unicentric Castleman disease
Idiopathic multicentric Castleman disease
KSHV/HHV8-associated multicentric Castleman disease

Campo E, Jaffe ES, Cook JR, et al. The International Consensus Classification of Mature Lymphoid Neoplasms: a report from the Clinical Advisory Committee. Blood 2022;140:1229-1253.

WHO Classification of Tumours Editorial Board. Haematolymphoid tumours. (WHO classification of tumours series, 5th ed.; vol. 11). Lyon (France): International Agency for Research on Cancer; 2024.

ABBREVIATIONS

C/A/P	chest/abdomen/pelvis
cART	combination antiretroviral therapy
CBC	complete blood count
CD	Castleman disease
CMV	cytomegalovirus
CRP	C-reactive protein
CTCAE	Common Terminology Criteria for Adverse Events
DLBCL	diffuse large B-cell lymphoma
EBER-ISH	Epstein-Barr virus-encoded RNA in situ hybridization
EBV	Epstein-Barr virus
ECOG	Eastern Cooperative Oncology Group
eGFR	estimated glomerular filtration rate
ESR	erythrocyte sedimentation rate
FDC	follicular dendritic cell
FDG	fluorodeoxyglucose
FLC	free light chain
FNA	fine-needle aspiration
GC	germinal center
GI	gastrointestinal
HHV8	human herpesvirus 8
HIV	human immunodeficiency virus
Ig	immunoglobulin
IGRT	image-guided radiation therapy
IHC	immunohistochemistry
iMCD	idiopathic multicentric Castleman disease
IMRT	intensity-modulated radiation therapy
IPL	idiopathic plasmacytic lymphadenopathy
ISRT	involved-site radiation therapy
IVF	in vitro fertilization
KS	Kaposi sarcoma
KSHV	Kaposi sarcoma-associated herpesvirus
LANA	latency-associated nuclear antigen
LDH	lactate dehydrogenase
mAb	monoclonal antibody
MCD	multicentric Castleman disease
MUGA	multigated acquisition
NGS	next-generation sequencing
NOS	not otherwise specified
PCR	polymerase chain reaction
POEMS	polyneuropathy, organomegaly, endocrinopathy, monoclonal protein, skin changes
SIFE	serum immunofixation electrophoresis
SPEP	serum protein electrophoresis
TAFRO	thrombocytopenia, anasarca, fever, reticulin fibrosis, and organomegaly
UCD	unicentric Castleman disease
VEGF	vascular endothelial growth factor

NCCN Evidence Blocks™

NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN Categories of Preference	
Preferred	Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other recommended	Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain circumstances	Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

This discussion corresponds to the NCCN Guidelines for Castleman Disease. Last updated: April 28, 2026.

Overview

Castleman disease (CD) is a relatively uncommon heterogenous group of lymphoproliferative disorders with an annual incidence of approximately 3336 to 5282 cases in the United States.[1] These disorders share common histopathologic features, and certain subtypes are associated with an increased risk of developing diffuse large B-cell lymphoma (DLBCL), Kaposi sarcoma (KS), and POEMS (polyneuropathy, organomegaly, endocrinopathy, monoclonal protein, skin changes) syndrome.[2-4]

Unicentric Castleman disease (UCD) and multicentric Castleman disease (MCD) are the two main clinical subtypes. In the 2022 newly revised WHO classification of hematolymphoid tumors (WHO5), UCD and MCD are classified as tumour-like lesions with B-cell predominance.[5] In the 2022 International Consensus Classification, UCD is not included and MCD is classified as human herpesvirus 8 (HHV8)-associated lymphoproliferative disorder.[6]

MCD is further divided into HHV8-positive (also known as KS herpesvirus [KSHV]-positive), HHV8-negative (also known as idiopathic MCD [iMCD]), and POEMS-MCD.[5] KSHV/HHV8-positive MCD is most commonly diagnosed in people with HIV (PWH) or otherwise immunocompromised individuals, and it has been reported as a precursor to aggressive B-cell lymphomas among PWH.[7,8]

iMCD is subclassified into three clinicopathologic subgroups: 1) iMCD-TAFRO (defined by thrombocytopenia [T], anasarca [A], fever [F], reticulin fibrosis of the bone marrow [R], and organomegaly [O] but generally has normal γ-globulin levels); 2) iMCD-not otherwise specified (NOS), which is typically characterized by hypergammaglobulinemia and thrombocytosis; and 3) idiopathic plasmacytic lymphadenopathy (IPL), a subgroup with distinct clinical features, including thrombocytosis, polyclonal hypergammaglobulinemia, and less severe fluid accumulation.[9-12]

The NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Castleman Disease provide recommendations for the diagnostic workup and treatment for UCD and MCD.

Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Guidelines are available at www.NCCN.org.

Sensitive/Inclusive Language Usage

The NCCN Guidelines® strive to use language that advances the goals of equity, inclusion, and representation. NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight-biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms men, women, female, and male when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to predominantly represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

Literature Search Criteria

Prior to the initial development of the NCCN Guidelines for Castleman Disease, an electronic search of the PubMed database was performed to obtain key literature in Castleman disease. The PubMed database was chosen because it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.[13]

The search results were narrowed by selecting relevant studies in humans published in English. The data from key PubMed articles and articles from additional sources deemed as relevant to these guidelines and discussed by the Panel have been included in this version of the Discussion section (eg, e-publications ahead of print, meeting abstracts). Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

Clinical Presentation

The clinical presentation of CD is often relatively nonspecific with enlarged lymph nodes, lymphadenopathy, and systemic symptoms.

UCD is characterized by involvement of a single lymph node or lymph node station and usually the absence of systemic symptoms.[14,15] This subtype generally has an indolent disease course with an excellent prognosis. UCD is most commonly (but not always) the histopathologic hyaline vascular variant and is nearly always HIV negative and HHV8 negative.[7,12] While often asymptomatic, some patients may experience compression symptoms due to compression of vital organs or other structures.[16-19] In a series of 404 patients with UCD, the most common sites of involvement were the mediastinum (29%), neck (23%), abdomen (21%), and retroperitoneum (17%).[16] In another analysis of 74 patients (43 patients with UCD and 31 patients with iMCD), systemic inflammatory symptoms, elevated inflammatory factors, and abnormal bone marrow features were more common in patients with iMCD than UCD.[17]

MCD is a remitting-relapsing disease with a variable natural history ranging from indolent disease with a very slow progression to acute and fulminant disease.[3,20] HHV8-positive MCD follows a more aggressive course. Immunocompromising conditions are a primary risk factor for HHV8-positive MCD, with HIV being the most common condition.[21] HHV8-positive MCD is more commonly associated with systemic symptoms such as fluid accumulation, cytopenias, liver and kidney dysfunction, and constitutional symptoms driven by cytokines such as interleukin 6 (IL-6).[17,20,22]

Individuals with IPL typically present with clinical features similar to MCD, including generalized superficial lymphadenopathy and constitutional symptoms; however, patients have less severe fluid accumulation (pleural effusion, ascites).[23] Other clinicopathologic features of IPL include significant plasmacytosis, hyperplastic germinal centers, thrombocytosis, and hypergammaglobulinemia. Although individuals with IPL present with a higher inflammatory state, they typically have a more indolent disease and better outcomes than those with other subtypes of iMCD, regardless of severity.[11]

iMCD affects multiple lymph node regions and exhibits lymphadenopathy in more than 1 lymph node or lymph node region. Clinical presentation can range from mild constitutional symptoms to life-threatening organ failure. iMCD-TAFRO generally presents with a more aggressive disease course and is associated with poorer outcomes than iMCD-NOS and IPL. The evaluation and determination of treatment should be similar to iMCD-NOS.[9-11]

Diagnosis

Due to the nonspecific and heterogenous nature of the disease, CD can often mimic other benign and malignant conditions.[2-4] Therefore, a high index of suspicion and histopathologic analysis is required for diagnosis of CD. Excisional or incisional biopsy of the lymph node (preferably the most fluorodeoxyglucose [FDG]-avid, accessible lymph node) is preferred for the definitive diagnosis of CD.[12,24-26]

A combination of clinical, imaging, and pathologic features should be used to establish the diagnosis of UCD, HHV8-positive MCD, and iMCD.[12,25] In an asymptomatic patient, histopathology consistent with CD along with the involvement of a single lymph node or region of lymph nodes can establish a diagnosis of UCD.[25,26]

There are three histopathologic subtypes across a spectrum from hyaline vascular to plasmacytic with mixed subtypes in between. All histopathologic subtypes can be observed across all subtypes of CD.[12,27] The hyaline vascular variant is most commonly associated with UCD, whereas the plasmacytic and mixed variants are more common in HHV8-associated MCD, and iMCD-NOS may exhibit any of the three histologic patterns. iMCD-TAFRO is more likely to have hyaline vascular-like histopathologic features, referred to as "hypervascular", or mixed subtypes.

The 2017 international diagnostic consensus criteria for the diagnosis of iMCD include two major criteria: histopathologic lymph node features consistent with the iMCD spectrum and enlarged lymph nodes in ≥2 lymph node stations; and 11 minor criteria:[12] anemia, thrombocytopenia, hypoalbuminemia, polyclonal hypergammaglobulinemia, renal dysfunction (elevated glomerular filtration rate [GFR] or proteinuria), elevated erythrocyte sedimentation rate (ESR) or C-reactive protein (CRP), constitutional symptoms, enlarged spleen and/or liver, fluid accumulation, eruptive cherry hemangiomatosis or violaceous papules, and lymphocytic interstitial pneumonitis.

The diagnosis of iMCD requires the presence of both major criteria, at least 2 of 11 minor criteria with at least 1 laboratory abnormality, and exclusion of infectious, malignant, and autoimmune disorders that can mimic iMCD.[12] See *Consensus Diagnostic Criteria for Idiopathic MCD* in the algorithm.

Adequate immunophenotyping of a surgically excised lymph node is required to confirm the diagnosis and exclude malignancy and other diseases.[12] The recommended immunohistochemistry (IHC) panel essential for diagnosis includes: kappa/lambda, CD20, CD3, and HHV8 latency-associated nuclear antigen (LANA). Additional IHC markers that may be useful under certain circumstances include Ki-67 index, immunoglobulin (Ig) heavy chains, CD5, CD10, BCL2, BCL6, cyclin D1, CD21, CD23, CD38, IRF4/MUM1, PAX5, and CD138. Epstein-Barr virus-encoded RNA in situ hybridization (EBER-ISH) and flow cytometry with peripheral blood and/or biopsy specimen of kappa/lambda, CD19, CD20, CD5, CD23, and CD10 may also be useful in certain circumstances. Other diagnostic testing that may be useful under certain circumstances includes molecular analysis to detect Ig gene rearrangements.

Workup

Initial workup should include a physical examination (with attention given to node-bearing areas, liver, and spleen) and evaluation of performance status. Laboratory assessments should include standard blood work including complete blood count (CBC) with differential, and a comprehensive metabolic panel. Serum lactate dehydrogenase (LDH), CRP, and ESR should be measured. Serum protein electrophoresis with serum immunofixation electrophoresis (SIFE) and/or quantitative Ig levels, and serum free light chain (FLC) assay should be evaluated if there is concern for POEMS syndrome. Pregnancy testing should be done in patients of childbearing potential if therapy is planned. Skull-base to mid-thigh FDG-PET/CT scan (preferred) or a chest/abdomen/pelvis CT with contrast of diagnostic quality are recommended as part of an initial diagnostic workup to confirm extent of disease and total nodal involvement.[24,26]

Testing to detect the presence of HIV, HHV8, and Epstein-Barr virus (EBV) polymerase chain reaction (PCR) should also be done as part of routine workup. HHV8-positive MCD is distinguished by positive testing for HHV8 in lymph node tissue as well as PCR positivity for HHV8 in plasma in addition to multiple affected lymph nodes and histopathology consistent with CD.[20] Screening for concurrent KS is recommended in the setting of HHV8 or HIV positivity since HHV8-positive MCD may occur with concomitant KS in ≤72% of patients.[28] Hepatitis B testing (if rituximab-containing regimens are being considered due to the risk of viral reactivation with anti-CD20 monoclonal antibody [mAb]-based regimens) and hepatitis C testing are also essential components of workup.

Bone marrow biopsy and aspirate as well as bone marrow biopsy with trichrome and reticulin stain (particularly in patients with TAFRO syndrome) may be useful under certain circumstances.[29,30] An echocardiogram or multigated acquisition (MUGA) scan can be considered if anthracycline-based regimens are being considered for treatment. Other components of workup that may be useful in certain circumstances include neck CT with contrast, IgG4, serum IL-6, serum IL-10, vascular endothelial growth factor (VEGF), uric acid, and ferritin. Discussion surrounding fertility preservation is also recommended as appropriate.

Treatment Recommendations

Unicentric Castleman Disease

UCD has an indolent disease course with an excellent prognosis. Careful consideration should be given to the risks and benefits of any therapy for patients with UCD, and observation can be considered in asymptomatic patients.[25]

First-Line Therapy

Surgical resection is the optimal treatment, wherever possible, as it has been associated with very low rates of recurrence and high relapse-free survival (RFS) in multiple case series and retrospective studies.[2,16,19,25,31-33] One case series of 53 patients reported a 5-year overall survival (OS) rate of 91%,[2] and another retrospective analysis of 278 patients showed an OS rate of >90% with up to 10-year follow-up.[16] Therefore, all patients with a diagnosis of UCD should be evaluated for resectability of disease.[25]

Observation until recurrence is recommended following complete resection. Patients who are asymptomatic following an incomplete resection may be observed until recurrence. Patients who remain symptomatic following incomplete resection should be treated as described below for unresectable disease.

Observation is also appropriate for asymptomatic patients with unresectable UCD. Radiation therapy (RT) or systemic therapy with the intent of eventual surgical resection are included as options for patients with unresectable disease. Available evidence (mostly from case reports) supports the use of involved-site RT (ISRT), or rituximab with or without steroids in patients with unresectable UCD and compressive symptoms.[25,34,35] Patients with inflammatory symptoms (eg, night sweats, fevers, weight loss) or laboratory disturbances (increased ESR or CRP) may have excessive cytokine production and may benefit from anti-IL-6 mAbs such as siltuximab and tocilizumab, although evidence in this setting is sparse. Therefore, tocilizumab or siltuximab are also included as options for HIV-1-negative/HHV8-negative, surgically unresectable UCD with primarily inflammatory symptoms. [25]

ISRT should be reserved for severe symptomatic and unresectable disease due to potential long-term complications.[25] The optimal dose of RT in unresectable UCD is not established. Doses as high as 40 Gy in 1.5 to 2 Gy fractions have been used.[34] Lower doses may be considered depending upon clinical circumstances and advanced radiation techniques are recommended to limit the radiation dose to surrounding normal tissues.[36] Patients with non-bulky disease may be observed after ISRT. Embolization may also be considered in certain situations to render surgical resection more feasible and reduce the risk of severe perioperative bleeding.[25]

Relapsed/Refractory Disease

Biopsy is encouraged in patients with relapsed/refractory disease to rule out transformation to DLBCL or concomitant development of other malignancies or opportunistic infections.

Treatment options for relapsed/refractory disease are the same as described above for unresectable disease: ISRT, embolization (if amenable), or systemic therapy with the intent of eventual surgical resection (rituximab ± prednisone or siltuximab [for HIV-negative and HHV8-negative disease], or tocilizumab [if there is a shortage of siltuximab or if it is not available]).[25]

Multicentric Castleman Disease

Treatment recommendations for MCD are based on the presence of criteria for active disease, fulminant/severe disease, or organ failure and HHV8 status. Patients may experience symptoms related to an excess of IL-6 production; establishing the severity of symptoms is a vital component of management, as severity can vary widely between patients. Corticosteroid monotherapy is not very effective for disease control but may be combined with other therapies for symptom control. Due to the rarity of the condition, available evidence for most of the treatment options is based on case reports making comparison between treatment options difficult.[24,37,38]

Patients with POEMS-associated MCD should have treatment directed at POEMS syndrome with regimens recommended for the treatment of multiple myeloma. High-dose therapy followed by autologous stem cell rescue (HDT/ASCR) may also result in favorable outcomes, particularly for patients with MCD associated with POEMS.[39,40]

First-Line Therapy

Criteria for Active Disease Present with No Organ Failure

In patients with criteria for active MCD (fever, increased serum CRP level >20 mg/L in the absence of any other etiology, and at least three other MCD-related symptoms) with no organ failure, first-line therapy options are based on HHV8 status as described below.

Idiopathic MCD (non-severe; HIV-1 and HHV8 negative)

Rituximab- and cyclophosphamide-containing regimens have demonstrated activity in patients with iMCD.[41-43] A systematic review of 47 patients treated with cytotoxic chemotherapy, with or without rituximab, reported a complete response (CR) rate of 44% with 93% of patients alive at 2 years.[41] In a retrospective study of 27 patients with iMCD, rituximab- and cyclophosphamide-containing regimens resulted in an overall response rate (ORR) of 56% with a CR rate of 33%.[42]

Siltuximab is FDA approved for the treatment of patients with iMCD that is HIV-1 and HHV8 negative, based on the results of a double-blind, placebo-controlled, international trial.[44-46] In this trial 79 patients with iMCD were randomized to siltuximab (n = 53) or placebo (n = 26). Siltuximab resulted in significantly improved durable tumor and symptomatic response rate compared to placebo (34% vs. 0%; P = .0012).[44] Long-term follow-up data, after a median follow-up of 6 years, also established the efficacy and safety of siltuximab.[45] Siltuximab was also associated with significantly improved PFS compared to placebo (median PFS was not reached for siltuximab compared to 15 months for placebo; P = .0001).[46] Adverse events of grade ≥3 were reported in 60% of patients, with hypertension (13%), fatigue (8%), nausea (7%), neutropenia (7%), and vomiting (5%) being the most common adverse events.

It is worth noting that in this pivotal trial responses were predominantly found in disease with the plasmacytic or mixed histologic variants and were not seen in the hyaline vascular variant where siltuximab may have less clinical activity.[44] However, secondary analyses of clinical trial and real-world data demonstrated that siltuximab resulted in similar clinical benefit across histopathologic subtypes, confirming the efficacy of siltuximab in the hyaline vascular variant also.[47] Importantly, in the aforementioned clinical trial, placebo was used as the control arm instead of an active treatment regimen, so it is unknown whether siltuximab is superior to chemoimmunotherapy. Therefore, chemoimmunotherapy can be considered as an appropriate treatment option in selected patients, particularly for patients with fulminant/severe disease.[41]

Rituximab (with or without prednisone), CVP (cyclophosphamide, vincristine, and prednisone) + rituximab, siltuximab, or tocilizumab are included as options for first-line therapy. Patients with disease responsive to rituximab-based time-limited therapy should be observed. Unlike chemoimmunotherapy, which is a fixed-duration treatment option, siltuximab is given continuously every 3 weeks by intravenous infusion in the absence of disease progression or intolerance. In patients with disease responding to siltuximab, treatment should be continued until disease progression, as disease may relapse on discontinuation of therapy.[24]

HHV8-Positive MCD (HIV-1-negative or HIV-1-positive)
Rituximab monotherapy is included as a first-line treatment option, and the strongest evidence for rituximab monotherapy comes from two phase II studies.[48,49] Rituximab has also been shown to decrease the risk of lymphoma in patients with HIV-associated disease.[50] Combination antiretroviral therapy (cART) should be given to all patients with HIV-1–positive disease.

In one prospective study of rituximab in HIV-associated MCD, 24 patients with relapsed/refractory, chemotherapy-dependent disease were given 4 weekly doses of rituximab at 375 mg/m^2. Twenty-two of the 24 patients were able to become independent of chemotherapy, and more than three quarters remained free from symptoms and progression 2 years after treatment.[48] In another phase II study of 21 patients with plasmablastic MCD, 20 patients achieved remission of symptoms, and 14 (67%) achieved a radiologic response. The median follow-up was 12 months. The OS and disease-free survival (DFS) rates at 2 years were 95% and 79%, respectively. The main adverse effect was reactivation of KS.[49] In another retrospective analysis of 113 patients (48 had previously received rituximab and 60 were previously diagnosed with KS), treatment with rituximab was associated with a higher 5-year OS rate compared to treatment with chemotherapy alone (90% vs. 47%).[50]

Occult KS is prevalent in HIV-1/HHV8-positive MCD and may flare after treatment with rituximab or prednisone.[50] The use of rituximab in combination with liposomal doxorubicin may reduce the incidence of KS exacerbation and may be particularly useful for patients with HHV8-positive MCD with concurrent KS.[51,52]

Zidovudine with valganciclovir has also been shown to be active in HHV8-positive MCD.[52,53] In a study that included 20 patients with HHV8-MCD alone and 34 patients with HHV8-MCD and KS, the 5-year PFS rate was 26% for treatment with the combination of zidovudine and valganciclovir.[52] The median PFS was 6 months. In a pilot study of 14 patients with HHV8-positive MCD, high-dose zidovudine in combination with valganciclovir resulted in major clinical and biochemical responses being attained in 86% and 50% of patients, respectively.[53]

Rituximab + prednisone, rituximab + liposomal doxorubicin (with or without prednisone), and zidovudine with either ganciclovir or valganciclovir are also included as options for first-line therapy.

Fulminant or Severe MCD With or Without Organ Failure
Combination chemotherapy regimens (CHOP [cyclophosphamide, doxorubicin, vincristine, and prednisone], CVAD [cyclophosphamide, vincristine, doxorubicin, and dexamethasone], and CVP with rituximab or

liposomal doxorubicin with rituximab [with or without prednisone]) are included as options for first-line therapy for HHV8-positive MCD. Combination chemotherapy may result in durable remissions.[24,34,37,38,54] Liposomal doxorubicin or rituximab monotherapy are appropriate treatment options for patients who are not candidates for combination chemotherapy.[48,49,51]

Siltuximab is a preferred first-line treatment option for HHV8-negative disease.[24] Single-agent tocilizumab or combination chemotherapy (CHOP, CVAD, and CVP) with or without rituximab are included as other recommended first-line therapy options for HHV8-negative iMCD.[24,34,37,38,54] Rituximab monotherapy is also included as an other recommended treatment option for patients who are not candidates for combination chemotherapy.

Relapsed/Refractory or Progressive Disease (criteria for active disease present or fulminant/severe disease)

Biopsy is encouraged in patients with relapsed/refractory disease to rule out transformation to DLBCL or concomitant development of other malignancies or opportunistic infections.

For active disease present with no organ failure, treatment with alternative first-line therapy regimens is recommended before initiating second-line and subsequent therapy (discussed below). Alternative single-agent or combination therapy should be considered for disease progression following second-line treatment options for relapsed/refractory disease. Treatment with an alternate second-line combination therapy regimen (with or without rituximab) is recommended for fulminant/severe disease with organ failure, prior to initiating alternative regimens for subsequent therapy (discussed below). Single-agent therapy (with ganciclovir or valganciclovir for HHV8-positive disease) is preferred for those with no organ failure, and combination chemotherapy (with or without rituximab) is preferred for patients with fulminant/severe disease and organ failure.

Etoposide (PO or IV), vinblastine, liposomal doxorubicin (for HHV8-positive disease), siltuximab (for HHV8-negative disease), or sirolimus (for HHV8-negative disease) are included as options for single-agent therapy. Tocilizumab is an appropriate option for second-line and subsequent therapy, if there is a shortage of siltuximab or if it is not available. CHOP, CVAD, and CVP (with or without rituximab) are the chemotherapy/chemoimmunotherapy options recommended for second-line therapy.[24,37,38] Liposomal doxorubicin with or without rituximab is included as an option for HHV8-positive disease with organ failure. Observation or maintenance with valganciclovir may be considered for HHV8-positive disease responding to treatment.

A prespecified analysis of the patients enrolled in the pivotal trial also showed that siltuximab is equally effective in patients with newly diagnosed as well as those with previously treated MCD (29 patients with previously treated iMCD in the siltuximab group and 17 patients in the placebo group). Siltuximab is included as an option for second-line therapy in patients with HHV8-negative disease.[55] Patients who received chemoimmunotherapy as initial treatment may receive siltuximab if needed subsequently for relapsed/refractory disease. Tocilizumab is an appropriate option, if there is a shortage of siltuximab or if it is not available.

The results of the retrospective study (n = 26 patients) showed that the mTOR inhibitor sirolimus was a well-tolerated and effective treatment for patients with relapsed/refractory iMCD.[56] After a median follow-up of 12 months, sirolimus resulted in a response rate of 69% (biologic and symptomatic response) and the median time to next treatment was 46 months. Sirolimus is included as a single-agent therapy option for second-line therapy in patients with HHV8-negative disease.

Alternative Regimens for Subsequent Therapy

Tocilizumab is an effective treatment option for patients with MCD that is refractory to rituximab and chemoimmunotherapy.[39,57-59]

Other options include bortezomib ± rituximab, thalidomide ± rituximab, lenalidomide ± rituximab, high-dose zidovudine + ganciclovir or valganciclovir (if HHV8-positive), and anakinra (IL-1 receptor antagonist), based on case reports and case series.[24,52,53,60-65]

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